Example: Interval Estimation

To estimate the average tensile strength of steel wires,
30 samples were tested. Find the 95% confidence interval.

Solution: 62.5 plus or minus 1.343

Context window

The model reads only the tokens inside its context window,
so the agent summarizes long files before editing.

Prompting

A prompt gives the model context, examples and the task,
and the agent plans before it edits the file.

Exercise 1

Write a prompt that makes the agent plan before editing.
Explain why the context window limits the plan.

Q1. What does the agent do first?

Choose one: plan, edit, summarize.

문제 2

Describe the context window in one sentence.

Agents

An agent loops over tools until the task is done, for example example, the agent plans, edits the file and runs the tests.

Evaluation

Tests and review catch errors that the model makes, and the API key stays outside the repository.

Gradient Descent

Gradient descent minimizes the loss function iteratively.
Example: learning rate controls the step size

Momentum accelerates convergence on flat regions.

Backpropagation

Backpropagation applies the chain rule layer by layer,
and the optimizer updates every weight with its gradient.